

TECHNOLOGY AT YOUR SERVICE

The world's first solar-powered aircraft takes flight

The world's first solar-powered aircraft, Solar Impulse 2, took off from Abu Dhabi, UAE, on its maiden journey around the world. The plane was built in Switzerland and took off from Al Bateen airport with the company's founder, André Borschberg, along with the company's co-founder Bertrand Piccard on board.



Solar-powered Solar Impulse 2

Project developers have said that the aim behind this airplane is to stress on the importance of giving up polluting technologies and opt for greener and clean technologies in aviation. Solar Impulse 2 features 17,428 photovoltaic cells that can capture the sun's energy. These cells can charge the four batteries that power the electric engines responsible for the thrust of the aircraft. The wings are 72m long and the aircraft weighs 2300kg.

Two pilots will pilot the aircraft taking turns during the 35,000km flight and will make stops at Oman, India, Myanmar and China, followed by crossing the Pacific and reaching Hawaii, the USA. The plane will then cross the Atlantic to enter Europe and Morocco before the trip comes to an end in Abu Dhabi.

The plane can reach altitudes of 8500m during the day to capture sun's rays, and during night it can drop down to 1500m while flying over the ocean.

A 3D-printed open source Raspberry Pi-powered robot

3D printing technology is being used in our lives in amazing ways. Desktop 3D printing is a common trend now and is slowly heading towards convergence between 3D printing and robotics. The credit-card-sized

Raspberry Pi has been of great help, along with the affordability factor of desktop 3D printers. Now, with the help of these two, people can create their own robotic devices. That is how Pi Tank, the robot, was born.

Pi Tank is a product by Madis Kaasik and his team, and is a result of 3D printing. It was a challenge for the team to come up with something unique, and that is when they came up with a 3D-printable Raspberry Pi open source tank robot that utilises a Microsoft Kinect sensor. Pi Tank took inspiration from Parrot Drone app and would allow users to easily see what the robot is seeing at any given time.

After completing the design work, the team started with 3D printing the robot using Mass Portal Pharaoh Ed 3D printer, which was provided to all participants of Garage48 event, where the team was also participating.

Microsoft Kinect allows 3D scanning of objects too. The robot is powered by four geared motors, Raspberry Pi 2, Wi-Fi dongle, Arduino Uno, 12V-to-USB board, S3 2200 LiPo battery and L298n motor driver board.

Pi Tank aims to create a robot that can do whatever you want it to. Developers behind



Pi Tank robot

the project are Ilya Kuzovkin (programmer and app developer), Ulf Anso (programmer, electronics and mechanical engineer), Kristjan Jarvan (team lead, marketing) and Madis Kaasik (mechanical engineering, design, CAD and 3D printing).

Objet260 dental-selection 3D printer takes digital dentistry to the next level

The versatile new 3D printer is designed to help mid- to large-sized dental and orthodontic labs grow their businesses by producing realistic models with true-to-life